Greecon
SH.P.K

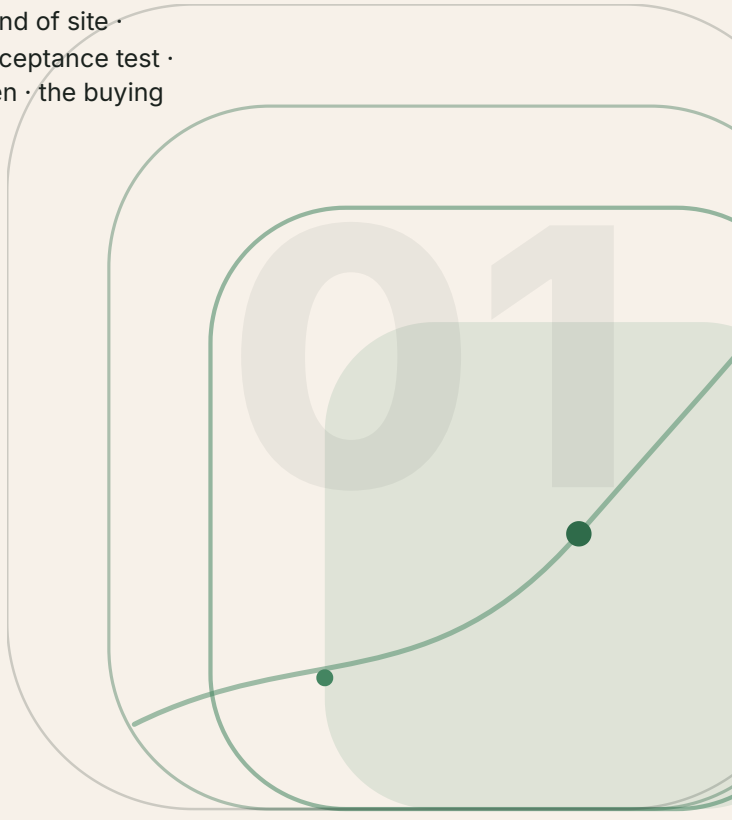
01

HARDWARE SHEET HW01 · QUALIFIED BY SPECIFICATION

The box

The small computer at the site. It reads every device, sends the readings to Greecon, and keeps them while the internet is down.

QUALIFIES	Qualified by specification. Any small computer that meets the specification on this sheet. What it must do well is come back by itself after a power cut.
USED IN	Kits H, S, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys, prepares and fits the box, and the founder who approves the purchase
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW01

The box

The box is the small computer at the site. About every ten seconds it asks each device for its reading. It sends the readings to Greecon over the internet, keeps them while the internet is down, and says every 30 seconds that it is alive. It is also what looks for the equipment when a person presses **Find equipment**. It runs one small program and needs very little: what is hard is coming back after a power cut, and that is what its specification is about.

At a glance

WHAT IT DOES	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.
HOW IT QUALIFIES	**Qualified by specification.** Any small computer that meets the specification on this sheet. What it must do well is come back by itself after a power cut.
SPECIFICATION	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input
FITTED BY	The project manager: prepared and tested at the office, plugged in on site.
PLANNING ESTIMATE (EUR)	180 to 300. Not a quote.
USED IN	Kits H, S, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

Any make qualifies that meets every line below. Nothing about a box is special, so a failed one can be replaced by anyone, with any box that qualifies.

It qualifies if:

- ☐ It runs Linux on a processor of the x86-64, 64-bit ARM or 32-bit ARM kind: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS (64-bit), with systemd. The installer checks this.
- ☐ It has 4 GB of memory or more and 32 GB of storage or more. That is the least the software runs on; the kit list buys 8 GB and 128 GB.
- ☐ Its storage is a real disk: an SSD or eMMC. Not an SD card.
- ☐ It has a real-time clock with a battery, and a BIOS setting of “power on after power loss” (often “Restore on AC Power Loss: Power On”).
- ☐ It takes 9 to 36 V DC, or a plain 12 V, so that it can sit behind a DC UPS.
- ☐ It has no fan, and works from 0 to 50 °C or better: a plant room is hot and dusty.

- [ ] It has one Ethernet port. Two is nicer.

Option	When	Planning estimate (EUR)	Note
Fanless x86 mini PC (an Intel N100-class machine)	The default. Most sites	180 to 300	Check the clock battery and the power-on setting in the BIOS before buying many
Raspberry Pi 5 (4 or 8 GB)	A bench box, or a small site, with a proper SSD on NVMe or USB 3 and a clock battery. Never an SD card	120 to 200 with SSD and case	Starts by itself when power returns
Industrial PC (Advantech UNO, OnLogic, Seeed reTerminal DM class)	Hot, dusty or vibrating sites; customers who insist on industrial parts	450 to 900	The same software runs on it. Not needed to begin

STOP

Do not buy a box that starts from an SD card: it is the commonest reason these boxes fail, because an SD card corrupts on a power cut and wears out under constant writing. Do not buy one that has no clock battery, or that stays off when the power returns. Shop listings rarely say whether there is a clock battery: ask the vendor, in writing.

At each kind of site

Every site has one box. What changes is where it sits and what it reads.

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	In the plant room. It reads the grid meter, the solar meter, the water module and the pump contact module: four devices, seven readings.
S Solar plant with grid connection	IN THE KIT	1	Near the main board or the inverter. It reads the grid meter and the solar meter and, where they are fitted, the load meter and the battery.
F Farm or greenhouse with irrigation	IN THE KIT	1	In the pump house. It reads the soil probes, the weather sensor, the water module and, if there is a pump to watch, the pump contact module. A 4G router is the likelier way to the internet.
W Water tank and pump station	IN THE KIT	1	In the pump house or the plant room. It reads the water module, the pump contact module and, if one is fitted, the pump meter.

Wherever it sits, the place is dry, has 30 cm free around the box, is away from heat, and is within reach of a socket that stays on and of a network cable.

What it does by itself

- It reads every device about every ten seconds and sends what it reads.

- It keeps about three days of readings if the internet is down, for a site of up to about thirteen devices (a site with more devices keeps fewer days), and sends them in order when the internet returns. In a long outage it keeps one reading a minute for each reading, so that three days fit.
- It reports every 30 seconds, so that Greecon knows it is alive. Greecon's own team is told when a customer's box has been silent for a quarter of an hour.
- It takes new versions of its software from the platform, and goes back to the version before if a new one fails to start three times.
- It restarts if it fails.
- It only reads. It never writes to a meter or a module. Nothing on the internet can connect to it: it only makes outgoing connections.

Settings, and where they are made

Everything is set at the office, on a desk, before the kit leaves. It takes about twenty minutes, then the install line.

Setting	What to do
The operating system	Debian 12 from the "netinst" image, with only "SSH server" and "standard system utilities". No desktop. One user with a long password, kept in the password manager against the site's name. Host name `greecon-<site>`
The clock	Time zone Europe/Tirane, and `timedatectl set-ntp true`
Security updates	`apt-get install -y unattended-upgrades`, and answer yes when it asks
The BIOS	Power on after power loss; no sleep; start from the disk only
The Greecon software	Paste the install line from the site's setup page (see *In the platform*)

What the installer changes, so that nobody wonders: `/opt/greecon` (the software), `/etc/greecon` (the box's credential, readable by the service only), `/var/lib/greecon-edge` (what it remembers), one service called `greecon-edge`, one user called `greecon`, and one command, `greecon-edge`. Nothing else.

Fitting

PROJECT MANAGER

- [ ] **Place:** in the enclosure if there is room, otherwise on a shelf beside it.
- [ ] **Power:** behind the UPS. A box that takes 9 to 36 V can share the 24 V supply; a box that takes 12 V keeps its own supply.
- [ ] **Network:** a cable to the customer's router or switch, or to the 4G router. Never a guest network. The box needs only outgoing HTTPS (port 443) to `api.greecon.earth` (and to `nodejs.org` when it is installed) and a time service (UDP 123). Nothing needs to come in.
- [ ] **Label** it with the site name and the date. Never write a code or a password on the box.

A box that was installed at the office keeps its credential when it is moved: unplug it, carry it to the site and plug it in, and it reports from the new place within a minute, with nothing typed again.

In the platform

The box is step 3 of the setup list, **Install the box**.

1. On the site's setup page press **Add the box and get its install command**. A line appears that begins `curl -fsSL https://api...` and ends in a code like `GRC-XXXX-XXXX-XXXX`. It is shown once. The code works once, for seven days. Copy it and press **I have copied it**.
2. Open a terminal on the box (a keyboard and screen, or SSH) and paste the line exactly. It runs in seven numbered steps and finishes with `Done. Greecon can see this box.`
3. Refresh the setup page. The step says **Done**, with the box's name, its version and "Heard from N seconds ago". Nobody pressed anything to make it so: the platform saw the box.

STOP

The install line is the box's key. Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** for a new one.

To replace a box (failed, stolen or moved): press **Install a box, or replace this one** and confirm. A new line is shown, once. Install the operating system on the new box and paste the line. The new box takes the old one's place, and the old one's credential stops working at once. Nothing else is entered again: the addresses and the registers belong to the site, not to the box.

Acceptance test

Before a box leaves the office:

- ☐ The install line ran to `Done. Greecon can see this box.`
- ☐ The setup page shows the step done, with the box's name and "Heard from N seconds ago".
- ☐ `greecon-edge status` says when the platform last heard from the box, and does not say NOT REPORTING.
- ☐ The power was pulled and restored: the box started by itself.
- ☐ The box is labelled with the site name and the date.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). Cut only the site's internet for ten minutes: unplug the router's uplink, or take the SIM out of the 4G router. Leave the box and the converter plugged into the router.	`greecon-edge status` shows a queue while it is out; the queue is empty a few minutes after the internet returns, and the readings from the gap are in the downloads			☐ Pass ☐ Fail

The commands a project manager uses on the box:

Command	What it does
`greecon-edge status`	How the box is: heard by Greecon, each device, the queue
`journalctl -u greecon-edge -n 50 --no-pager`	What the box has been saying
`greecon-edge find`	Looks at the network and says what answers. It only reads
`sudo systemctl restart greecon-edge`	Restarts the box's software

What goes wrong

What you see	Likely cause	What to do
The install line stops with `STOP:`	The sentence says which: no internet, a firewall that blocks outgoing HTTPS on port 443, a clock so far out that secure connections fail, or a code already used or older than seven days	Do what it says. If it stopped before step 5, *Giving this box its credential*, run the same line again. From step 5 on the code is used up: run the line again with no code at the end, which repairs the box and keeps its credential
The box says Offline	A power cut at the site (the commonest cause, and it ends by itself), the site's internet, or the box	Phone the site: is the power on, is the router's light on? Then, on the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. The last lines say what is wrong
The box's log says the platform does not recognise this box's credential	The box was replaced, or its code was used for another box	Press **Install a box, or replace this one** and run the new line on the box
The setup page says its clock is N minutes fast	The box has no time service	`sudo timedatectl set-ntp true`
Readings are missing after a long outage	The queue holds about three days. It is for outages, not for a site that is never connected	Nothing is lost if the internet returns within that time. A site with no internet at all is not sold

More is in the runbooks (Project Manager Field Manual, PDF 02): R1 (a box is offline) and R10 (a box is replaced).

Not yet proven

- The heartbeat, the three-day queue and the updates with their way back are tested in software. They have not yet been shown on a real box with the internet really cut: that is why the internet-cut test is done at the first install, and its result written down.
- Installing the operating system and pasting the install line are still done in a terminal. There is not yet a box that sets itself up the first time it is switched on.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	